

Lab Session 1

The Rocq Proof Assistant

Software Verification

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The main purpose of this lab session is for you to get the first feeling of the Rocq proof assistant. The exercises from this lab session are inspired by the first chapter of *Software Foundations, Volume 1*: <https://softwarefoundations.cis.upenn.edu/1f-current/Basics.html>. I strongly recommend you to read this chapter and go through the remaining exercises.

1 Proof by Simplification

1.1 Boolean Values

Consider the following definition of Boolean values in Rocq:

```
Inductive bool : Type :=
| true
| false. Rocq
```

Exercise 1. Give definitions for the following functions:

- `negb`, Boolean negation
- `andb`, Boolean conjunction
- `orb`, Boolean disjunction

□

Exercise 2. Prove the following lemma:

```
Lemma unfold_andb: ∀ b1 b2: bool,
  andb b1 b2 = if b1 then b2 else false. Rocq
```

□

Exercise 3. Prove the following lemma:

```
Lemma andb_true_b: ∀ b: bool,
  andb true b = b. Rocq
```

□

Exercise 4. Prove the following lemma:

```
Lemma andb_false_b: ∀ b: bool,  
  andb false b = false.
```

Rocq

□

1.2 Natural Numbers

Consider the following definition of natural numbers in Rocq:

```
Module NatPlayground.  
  Inductive nat : Type :=  
    | 0  
    | S (n: nat)
```

Rocq

Exercise 5. Give definitions for the following functions:

- `plus`, addition of two natural numbers
- `mult`, multiplication of two natural numbers

□

Exercise 6. Prove the following lemma:

```
Lemma plus_0_n: ∀ n: nat,  
  plus 0 n = n.
```

Rocq

□

Exercise 7. Prove the following lemma:

```
Lemma mult_0_1: ∀ n: nat,  
  mult 0 n = 0.
```

Rocq

□

2 Proof by Rewriting

Exercise 8. Prove the following lemma:

```
Lemma plus_id: ∀ n m: nat,  
  n = m →  
  plus n n = plus m m.  
End NatPlayground.
```

Rocq

□

Exercise 9. Prove the following lemma:

```
Lemma mult_n_0_m_0: ∀ p q: nat,  
  plus (mult p 0) (mult q 0) = 0.
```

Rocq

□

Exercise 10. Prove the following lemma:

Lemma `mult_n_1`: $\forall n: \text{nat},$
`mult n (S 0) = n.`

Rocq

□

3 Proof by Case Analysis

3.1 Boolean Values

Exercise 11. Prove the following lemma:

Lemma `andb_b_false`: $\forall b: \text{bool},$
`andb b false = false.`

Rocq

□

Exercise 12. Prove the following lemma:

Lemma `negb_involutive`: $\forall b: \text{bool},$
`negb (negb b) = b.`

Rocq

□

Exercise 13. Prove the following lemma:

Lemma `andb_commutative`: $\forall b c: \text{bool},$
`andb b c = andb c b.`

Rocq

□

Exercise 14. Prove the following lemma:

Lemma `andb_true_elim`: $\forall b c: \text{bool},$
`andb b c = true  $\rightarrow$  c = true.`

Rocq

□

Exercise 15. Prove the following lemma:

Lemma `students_favorite`: $\forall b: \text{bool},$
`b = if b then true else false.`

Rocq

□

3.2 Natural Numbers

Exercise 16. Give definitions for the following function:

- `eqb`, equality between two natural numbers

□

Exercise 17. Prove the following lemma:

Lemma `plus_1_neq_0`: $\forall n: \text{nat},$
`eqb (plus n (S 0)) 0 = false.`

Rocq

□